Analysis Qualifier Prep
Problem Set 1 (Rookie Mix)

Uttaran Choudhurry

June 2026

Question 1:

Determine, with justification, the limit

$$\lim_{n\to\infty}\int_0^\infty \frac{1-cos(nx)}{n^2x^2}dx$$

Question 2:

Let (X, M, μ) be a finite measure space. Let $f : X \to \mathbb{C}$ be a measurable function. Show that $f \in L^\infty(X, \mu)$ if and only if

$$\sup_p \Big\{ \|f\|_p : p \in [1, \infty) \Big\} < \infty$$

Question 3:

Let $f \in L_1([0, \infty))$ (with respect to the usual Lebesgue measure). Suppose that f is uniformly continuous. Prove that $f \in C_0([0, \infty))$.

Question 4:

Let $C([0, 1])$ be the space of continuous functions on $[0, 1]$. Prove that

$$\|f\| = \left(\int_{[0,1]} |f|^2 dm\right)^{1/2}$$

defines a norm on $C([0, 1])$. Is $C([0, 1])$ a Banach space with respect to this norm? Justify your answer.

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Problem Set 2

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Question 1:

Suppose f is a Lebesgue measurable function on $\mathbb{R}$ such that f and $x :\to xf(x)$ are both in $L^2(\mathbb{R})$. Prove that $f \in L^1(\mathbb{R})$

Question 2:

Suppose that $f : [1, \infty) \to \mathbb{C}$ is Lebesgue integrable and f is continuous at 1. Prove that

$$\lim_{n\to\infty} n \int_1^\infty \frac{f(x)}{x^n} dx = f(1).$$

Question 3:

Let $p \in (1, \infty)$ and let $f_1, f_2, ... : \mathbb{R} \to \mathbb{C}$ be L^p functions. Assume that $\sum_{n=1}^\infty \|f_n\|_p^p < \infty$.

1. Suppose that the sets $E_n = \{x \in \mathbb{R} : f_n(x) \neq 0\}$ are disjoint. Prove that the series $\sum_{n=1}^\infty f_n$ converges in $L^p(\mathbb{R})$.
2. Give an example to show that the conclusion of part (1) can fail if the sets E_n are not assumed disjoint.

Question 4:

For which $p \in [1, \infty]$ is there a nonzero bounded linear functional T on $L^p([0, 1])$ such that T vanishes on the subspace $C([0, 1]) \subset L^p([0, 1])$? Justify your answer.

Analysis Qualifier Prep
Problem Set 3

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July 2026

Question 1:

Let $\{f_n\}$ be a sequence of Lebesgue integrable functions on the real line. Suppose that $|f_n(x)| \leq 1$ for all $x \in \mathbf{R}$ and f_n converges to zero almost everywhere. Is it true that

$$\lim_{n\to\infty} \int_{\mathbf{R}} f_n dm = 0,$$

where m is the Lebesgue measure? It true explain else if false, explain what had to change.

Question 2:

Let $(X, \mathcal{M}, \mu)$ be a finite measure space and let $\{f_n\}$ be a sequence of measurable functions which converges to zero almost everywhere (with respect to μ). Prove that f_n converges to zero in measure.

Question 3:

Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of functions in $L^2(\mathbb{R})$. Suppose that for all $n \in \mathbb{N}$, the Fourier transform $\widehat{f_n}$ vanishes outside $[n, n+1]$ and that

$$\|f_n\|_2 = n^{-\frac{51}{100}}.$$

Show that the series $\sum_{n=1}^{\infty} f_n$ converges in norm in $L^2(\mathbb{R})$.

Question 4:

Let $f \in L^1(\mathbb{R})$ and $g = \chi_{[0,1]}$. Suppose that their convolution $f * g = 0$ for a.e $x \in \mathbb{R}$. Prove that $f = 0$ a.e.

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Problem Set 4

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Question 1:

Recall that l^∞ is the set of bounded sequences of real numbers. Define $c = \{\{x_n\} \in l^\infty, \lim_{n\to\infty} x_n \text{ exists}\}$ with the induced supremum norm. Show

(i) For $\{\eta_n\} \in l^1$ and $\alpha \in \mathbb{R}$, functional

$$f(\{x_n\}) = \sum_{n=1}^{\infty} \eta_n x_n + \alpha \cdot \lim_{m\to\infty} x_m$$

is a bounded linear functional (i.e., in $(c)^*$).

(ii) The norm of the functional f is given by

$$\|f\| = \|\{\eta_n\}\|_{l^1} + |\alpha|.$$

Question 2:

Let $T : X \to Y$ be bounded linear operator between normed spaces.

(a)Prove that for any $x, w \in X$ we have

$$\max\{\|T(x+w)\|, \|T(x-w)\|\} \geq \|T(x)\|. \tag{1}$$

(b)Prove that for any $x \in X$ and $r > 0$

$$\sup_{\tilde{x}\in B(x,r)} \|T(\tilde{x})\| \geq r\|T\|. \tag{2}$$

where $B(x, r)$ is the open ball in X centered at x with radius r.

Question 3:

Suppose that $\{f_n\}$ is a sequence of Lebesgue integrable function on $[0, 1]$ such that $|f_n(x)| \leq \pi$ for all n and $x \in [0, 1]$. Suppose that

$$\lim_{n\to\infty} f_n(x) = 0 \ a.e.m. \text{ and } \left|\frac{f_n(x)}{x^2}\right| \leq x^{1/2} \text{ for all } x \in (0, 1],\ a.e.m. \text{ and } n \in \mathbb{N}.$$

Compute the
$$\lim_{n\to\infty}\int_{(0,1]}\frac{f_n(x)}{1-\cos(x)}dm,$$
where m is the Lebesgue measure. Justify your computation.

Question 4:

Give an example of a sequence of functions with $\inf_n \|f_n\|_\infty \geq M$ for some $M > 0$ and a function f such that $\lim_{n\to\infty} \|f_n - f\|_p = 0$ for all $p > 0$, but for which $\|f\|_\infty < M$.